

March 11, 2026 Tutorial exercise 3

(Homework 1)

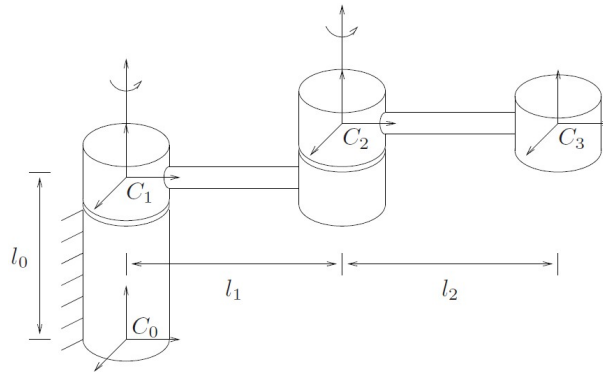


Figure 2.17: A two degree of freedom manipulator.

C_0 is an inertial axis system. Manipulator arm C_1 with body axis system C_1 hinges on base C_0 around the inertial z axis, which is also the C_1 z axis. Manipulator arm C_2 with body axis system C_2 hinges on C_1 around the C_2 z axis. The origin of the C_2 frame is located a distance l_1 from the origin of the C_1 frame, along the C_1 y axis. Point C_3 is located a distance l_2 from the origin of the C_2 frame, along the C_2 y axis. Even though the drawing clearly show otherwise, you can by approximation assume that the origins of the C_1 and C_2 frames and point C_3 lie in a single plane normal to the inertial z axis, a distance l_0 along this axis, from the inertial frame origin.

Use the following parameter values:

- $l_0 = 0.8$ m
- $l_1 = 0.6$ m
- $l_2 = 0.4$ m

All rotation angles about axes are considered positive as defined with a right hand rule, with the thumb pointing in the positive axis direction.

The figure above shows the robot in its initial configuration, with both arms aligned in a direction parallel to the inertial y axis. Frame C_1 is now rotated about its z axis through 60 degrees and frame C_2 is rotated about its z axis, relative to frame C_1 , through -135 degrees.

Use the homogeneous description of rigid motion on MATLAB and determine the coordinates of point C_3 in the inertial axis system, as well as the total angle through which the arm of C_2 has rotated in the inertial axis system.